

- Develop natural language processing applications with transformers and open datasets.

AI may be used as a tool to support your process with proper documentation but may **not** create or write any part of your project for you. AI-generated projects will result in an academic integrity violation as outlined in the Study.com Academic Integrity Policy.

Instructions

In this assignment, you are required to design and implement an FAQ Chatbot that responds to user questions about a specific product. The chatbot should use a Large Language Model (LLM), such as OpenAI GPT, Anthropic Claude, or similar, and the LangChain framework to manage prompts, context, and data retrieval.

You may select any consumer product, such as home appliances, grocery delivery, online retail, or streaming services. The chatbot's responses should be based on relevant and realistic content sourced from:

- Open datasets available on platforms like Kaggle or Google Dataset Search
- FAQs or product information from official company website

You must write the code in Python and utilize LangChain along with an LLM API (e.g., OpenAI API). You may incorporate commonly used libraries such as Pandas, ChromaDB, and FAISS.

Your submission must include two core components:

- source code (written in Python and provided in .py format)
- a report that follows the guidelines outlined below

Report Guidelines

Your report must include the following sections:

- Title Page
- Name and Purpose of the Chatbot
- NLP/LLM Methods Used
- Dataset Information
- Libraries, Toolkits, and Framework
- Application Design and Implementation
- Instructions for Running the Chatbot
- Results
- Discussion and Insights
- References

Sample Report Template

Developing an FAQ Chatbot Using LangChain and LLM APIs

1. Name and Purpose of the Chatbot

- Provide the name of your chatbot and explain its purpose in one paragraph.

2. NLP/LLM Methods Used

- Describe which NLP/LLM methods were applied and why they were chosen.

3. Dataset Information

- 3.1 Dataset source/link
- 3.2 Number of records
- 3.3 Number of features
- 3.4 Description of features: Feature Name, Description, Data Type
- 3.5 Preprocessing steps

4. Libraries, Toolkits, and Frameworks

- List all libraries used and explain their roles.

5. Application Design and Implementation

- Describe how the chatbot works, including data flow, model training, and prediction.

6. Instructions for Running the Chatbot

- Provide a step-by-step guide on how to execute the code. Include environment setup or dependencies.

7. Results

- Include screenshots that demonstrate the chatbot in operation and display the prediction results.

8. Discussion and Insights

- Reflect on your model's performance, limitations, and potential improvements.

9. References

- List all data sources, libraries, and references in APA style.

You may refer to the course material for supporting evidence, but you must also use **3 external sources**. All sources should be cited using APA format.

Formatting

Please submit your project as one of the following:

a .txt file

a link to your cloud storage, such as Google Drive, Dropbox, or OneDrive, containing your work. Please ensure the privacy settings allow access. For example, if you're using Google Drive, make sure the file is set to 'anyone with the link can view'.

Grading

Your project will be graded based on the following rubric:

Criteria	Excellent (5)	Good (4)	Needs Improvement (2-3)	Unacceptable (1)	Total Possible Points
Problem Definition & NLP/LLM Approach (x3)	Purpose of chatbot clearly described; strong explanation of NLP/LLM methods and why the chosen model/framework fits the task.	Clear purpose and appropriate NLP/LLM method with minor gaps in explanation.	Unclear purpose or limited explanation of chosen LLM/NLP methods.	No clear purpose described; no meaningful NLP/LLM explanation.	15
Dataset Selection & Preprocessing (x4)	Dataset thoroughly described (source, features, size); preprocessing steps (cleaning, embeddings, etc.) clearly justified and appropriate.	Dataset and preprocessing steps clearly described and appropriate.	Limited dataset description; minimal or unclear preprocessing.	No dataset provided, or dataset provided is incomplete or irrelevant; preprocessing steps not described or inappropriate.	20
Application Design & Implementation (Python + LangChain) (x3)	Well-organized, functional code; correct use of LangChain, vector store, and integration with LLM; architecture clearly explained.	Generally functional code with minor issues; explanation mostly clear.	Code partially functional or poorly structured; weak explanation.	Nonfunctional or missing code.	15

Chatbot Performance & Demonstrated Results (x3)	Clear evidence of chatbot operation (screenshots/logs); responses accurate and grounded in product data; retrieval works effectively.	Chatbot works with minor errors; results shown.	Chatbot only partially works; inconsistent retrieval; limited evidence of testing.	No working chatbot or no results shown.	15
Discussion, Insights & Improvement Plan (x2)	Thoughtful reflection on performance, limitations, improvements, and real-world applicability; demonstrates understanding of course concepts.	Clear reflection with some meaningful insights but may lack depth in parts.	Unclear or superficial reflection with limited insights.	Missing or irrelevant reflection with no meaningful insights	10

Grading Timeline & Policies:

- You should receive your project grade within 48 hours.
- If you are not satisfied with the score you receive, you may revise and resubmit for grading.
- Keep in mind that the grade you receive on your project is only a portion of your overall grade for the course. Please see the course syllabus for a more detailed breakdown of the grading policy.

Resources

Related Lessons

- Natural Language Processing & Deep Learning
- Practical Application for Artificial Intelligence: Learning Chatbot
- What is a Large Language Model (LLM)?
- Large Language Models and Artificial Intelligence

Writing Help

For additional guidance on completing this project, consult the following:

- Research Sources